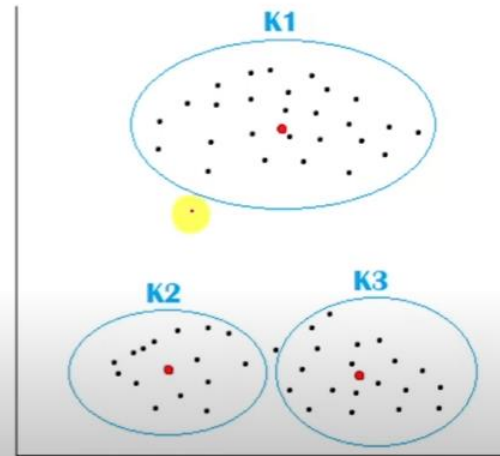


Fuzzy C-Means Clustering

Clustering

- Clustering is a **distance-based unsupervised machine learning algorithm** where data points that are close to each other are grouped in a given number of clusters/groups.



Hard Clustering

- In hard clustering each datapoint is assigned only a single cluster.
- The K-Means, K-Medoid clustering algorithms are hard clustering algorithms.

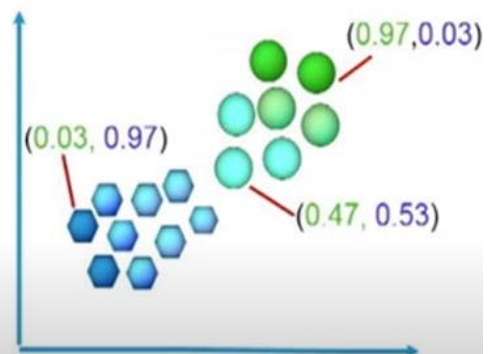
Hard Clustering



Soft Clustering

- On the other hand in soft clustering each data point belongs to a cluster with a certain probability also known as Membership Value.
- FCM (Fuzzy C-means clustering) algorithm is an example of soft clustering.

Soft Clustering



Fuzzy C-Means Clustering Steps

Step-1. Initialize membership table based on the number of clusters

Step-2. Find out the centroid

Step-3. Find out the distance of each point from centroids

Step-4. Update membership values

Step-5. Repeat step 2-4 until we get constant membership values or the difference is less than tolerance

Fuzzy C-Means Clustering – Steps

- **Step 1:** Given the data points based on the number of cluster required initialize the membership table with random values.
- Suppose the given data points are $\{(1, 3), (2, 5), (6, 8), (7, 9)\}$

Cluster	(1, 3)	(2, 5)	(4, 8)	(7, 9)
1	0.8	0.7	0.2	0.1
2	0.2	0.3	0.8	0.9

- **Step 2: Find out the centroid.**

- The formula for finding out the centroid (V) is:

- $$V_{ij} = \frac{\sum_{k=1}^n \gamma_{ik}^m * x_k}{\sum_{k=1}^n \gamma_{ik}^m}$$

- γ : Fuzzy membership value
- m : Fuzziness parameter generally taken as 2 and
- x_k is the data point

Cluster	(1, 3)	(2, 5)	(4, 8)	(7, 9)
1	0.8	0.7	0.2	0.1
2	0.2	0.3	0.8	0.9

- **Step 2: Find out the centroid.**

- The formula for finding out the centroid (V) is:

- $$V_{ij} = \frac{\sum_{k=1}^n \gamma_{ik}^m * x_k}{\sum_{k=1}^n \gamma_{ik}^m}$$

- γ : Fuzzy membership value
- m : Fuzziness parameter generally taken as 2 and
- x_k is the data point

- $$V_{11} = \frac{(0.8^2 * 1 + 0.7^2 * 2 + 0.2^2 * 4 + 0.1^2 * 7)}{(0.8^2 + 0.7^2 + 0.2^2 + 0.1^2)} = 1.568$$

- $$V_{12} = \frac{(0.8^2 * 3 + 0.7^2 * 5 + 0.2^2 * 8 + 0.1^2 * 9)}{(0.8^2 + 0.7^2 + 0.2^2 + 0.1^2)} = 4.051$$

Cluster	(1, 3)	(2, 5)	(4, 8)	(7, 9)
1	0.8	0.7	0.2	0.1
2	0.2	0.3	0.8	0.9

- **Step 2: Find out the centroid.**

- The formula for finding out the centroid (V) is:

- $$V_{ij} = \frac{\sum_{k=1}^n \gamma_{ik}^m * x_k}{\sum_{k=1}^n \gamma_{ik}^m}$$

- γ : Fuzzy membership value
- m : Fuzziness parameter generally taken as 2 and
- x_k is the data point

- $$V_{21} = \frac{(0.2^2 * 1 + 0.3^2 * 2 + 0.8^2 * 4 + 0.9^2 * 7)}{(0.2^2 + 0.3^2 + 0.8^2 + 0.9^2)} = \underline{5.35}$$

Cluster	(1, 3)	(2, 5)	(4, 8)	(7, 9)
1	0.8	0.7	0.2	0.1
2	0.2	0.3	0.8	0.9

- $$V_{22} = \frac{(0.2^2 * 3 + 0.3^2 * 5 + 0.8^2 * 8 + 0.9^2 * 9)}{(0.2^2 + 0.3^2 + 0.8^2 + 0.9^2)} = \underline{8.215}$$

Centroids are: (1.568, 4.051) and (5.35, 8.215)

• **Step 3: Find out the distance of each point from the centroid.**

- $D_{11} = \sqrt{(1 - 1.568)^2 + (3 - 4.051)^2} = 1.2$
- $D_{12} = \sqrt{(1 - 5.35)^2 + (3 - 8.215)^2} = 6.79$
- $D_{21} = \sqrt{(2 - 1.568)^2 + (5 - 4.051)^2} = 1.04$
- $D_{22} = \sqrt{(2 - 5.35)^2 + (5 - 8.215)^2} = 4.64$
- $D_{31} = \sqrt{(4 - 1.568)^2 + (8 - 4.051)^2} = 4.63$
- $D_{32} = \sqrt{(4 - 5.35)^2 + (8 - 8.215)^2} = 1.36$
- $D_{31} = \sqrt{(7 - 1.568)^2 + (9 - 4.051)^2} = 7.34$
- $D_{32} = \sqrt{(7 - 5.35)^2 + (9 - 8.215)^2} = 1.82$

Centroids are:

**(1.568, 4.051) and
(5.35, 8.215)**

Cluster	(1, 3)	(2, 5)	(4, 8)	(7, 9)
1	0.8	0.7	0.2	0.1
2	0.2	0.3	0.8	0.9
	1	1	2	2

- **Step 4: Updating membership values.**

- $$\gamma_{ki} = \left(\sum_{j=1}^n \left\{ \frac{d_{ki}^2}{d_{kj}^2} \right\}^{\left(\frac{1}{m-1} \right)} \right)^{-1}$$

- For point 1 new membership values are:

- $$\gamma_{11} = \left(\left\{ \frac{(1.2)^2}{(1.2)^2} + \frac{(1.2)^2}{(6.79)^2} \right\}^{\left(\frac{1}{(2-1)} \right)} \right)^{-1} = \underline{0.97}$$
- $$\gamma_{12} = \left(\left\{ \frac{(6.79)^2}{(1.2)^2} + \frac{(6.79)^2}{(6.79)^2} \right\}^{\left(\frac{1}{(2-1)} \right)} \right)^{-1} = 0.03$$

$$D_{11} = \underline{1.2}, \quad D_{12} = \underline{6.79}$$

$$D_{21} = 1.04, \quad D_{22} = 4.64$$

$$D_{31} = 4.63, \quad D_{32} = 1.36$$

$$D_{31} = 7.34, \quad D_{32} = 1.82$$

Cluster	(1, 3)	(2, 5)	(4, 8)	(7, 9)
1	0.8	0.7	0.2	0.1
2	0.2	0.3	0.8	0.9

k=data point

j=no of clusters

m=membership parameter

- **Step 4: Updating membership values.**

- $$\gamma_{ki} = \left(\sum_{j=1}^n \left\{ \frac{d_{ki}^2}{d_{kj}^2} \right\}^{\left(\frac{1}{m-1}\right)} \right)^{-1}$$

- For point 1 new membership values are:

- $$\gamma_{11} = \left(\left\{ \frac{(1.2)^2}{(1.2)^2} + \frac{(1.2)^2}{(6.79)^2} \right\}^{\left(\frac{1}{(2-1)}\right)} \right)^{-1} = 0.97$$

- $$\gamma_{12} = \left(\left\{ \frac{(6.79)^2}{(1.2)^2} + \frac{(6.79)^2}{(6.79)^2} \right\}^{\left(\frac{1}{(2-1)}\right)} \right)^{-1} = 0.03$$

$$D_{11} = 1.2, \quad D_{12} = 6.79$$

$$D_{21} = 1.04, \quad D_{22} = 4.64$$

$$D_{31} = 4.63, \quad D_{32} = 1.36$$

$$D_{31} = 7.34, \quad D_{32} = 1.82$$

Cluster	(1, 3)	(2, 5)	(4, 8)	(7, 9)
1	0.97	0.7	0.2	0.1
2	0.03	0.3	0.8	0.9

- **Step 4: Updating membership values.**

- $$\gamma_{ki} = \left(\sum_{j=1}^n \left\{ \frac{d_{ki}^2}{d_{kj}^2} \right\}^{\left(\frac{1}{(m-1)} \right)} \right)^{-1}$$

- For point 2 new membership values are:

- $$\gamma_{21} = \left(\left\{ \frac{(1.04)^2}{(1.04)^2} + \frac{(1.04)^2}{(4.64)^2} \right\}^{\left(\frac{1}{(2-1)} \right)} \right)^{-1} = 0.95$$

$$D_{11} = 1.2, \quad D_{12} = 6.79$$

$$D_{21} = 1.04, \quad D_{22} = 4.64$$

$$D_{31} = 4.63, \quad D_{32} = 1.36$$

$$D_{31} = 7.34, \quad D_{32} = 1.82$$

Cluster	(1, 3)	(2, 5)	(4, 8)	(7, 9)
1	0.97	0.7	0.2	0.1
2	0.03	0.3	0.8	0.9

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- **Step 4: Updating membership values.**

- $$\gamma_{ki} = \left(\sum_{j=1}^n \left\{ \frac{d_{ki}^2}{d_{kj}^2} \right\}^{\left(\frac{1}{(m-1)} \right)} \right)^{-1}$$

- For point 2 new membership values are:

- $$\gamma_{21} = \left(\left\{ \frac{(1.04)^2}{(1.04)^2} + \frac{(1.04)^2}{(4.64)^2} \right\}^{\left(\frac{1}{(2-1)} \right)} \right)^{-1} = 0.95$$

- $$\gamma_{22} = \left(\left\{ \frac{(4.64)^2}{(1.04)^2} + \frac{(4.64)^2}{(4.64)^2} \right\}^{\left(\frac{1}{(2-1)} \right)} \right)^{-1} = 0.05$$

$$D_{11} = 1.2, \quad D_{12} = 6.79$$

$$D_{21} = 1.04, \quad D_{22} = 4.64$$

$$D_{31} = 4.63, \quad D_{32} = 1.36$$

$$D_{31} = 7.34, \quad D_{32} = 1.82$$

Cluster	(1, 3)	(2, 5)	(4, 8)	(7, 9)
1	0.97	0.95	0.2	0.1
2	0.03	0.05	0.8	0.9

- **Step 4: Updating membership values.**

- $$\gamma_{ki} = \left(\sum_{j=1}^n \left\{ \frac{d_{ki}^2}{d_{kj}^2} \right\}^{\left(\frac{1}{(m-1)} \right)} \right)^{-1}$$

- For point 3 new membership values are:

- $$\gamma_{31} = \left(\left\{ \frac{(4.63)^2}{(4.63)^2} + \frac{(4.63)^2}{(1.36)^2} \right\}^{\left(\frac{1}{(2-1)} \right)} \right)^{-1} = 0.08$$

- $$\gamma_{32} = \left(\left\{ \frac{(1.36)^2}{(4.63)^2} + \frac{(1.36)^2}{(1.36)^2} \right\}^{\left(\frac{1}{(2-1)} \right)} \right)^{-1} = 0.92$$

$$D_{11} = 1.2, \quad D_{12} = 6.79$$

$$D_{21} = 1.04, \quad D_{22} = 4.64$$

$$D_{31} = 4.63, \quad D_{32} = 1.36$$

$$D_{31} = 7.34, \quad D_{32} = 1.82$$

Cluster	(1, 3)	(2, 5)	(4, 8)	(7, 9)
1	0.97	0.95	0.08	0.1
2	0.03	0.05	0.92	0.9

Cluster	(1, 3)	(2, 5)	(4, 8)	(7, 9)
1	0.8	0.7	0.2	0.1
2	0.2	0.3	0.8	0.9

Cluster	(1, 3)	(2, 5)	(4, 8)	(7, 9)
1	0.97	0.95	0.08	0.06
2	0.03	0.05	0.92	0.94

- **Step 5:** Repeat the steps (2-4) until the constant values are obtained for the membership values or the difference is less than the tolerance value

Cluster	(1, 3)	(2, 5)	(4, 8)	(7, 9)
1	0.8	0.7	0.2	0.1
2	0.2	0.3	0.8	0.9

Cluster	(1, 3)	(2, 5)	(4, 8)	(7, 9)
1	0.97	0.95	0.08	0.06
2	0.03	0.05	0.92	0.94

$$+ = 0.01$$

Practice Question:

Apply FCM algorithm to generate the clusters from the dataset $\{(1,1),(1,2),(2,2),(3,2)\}$